Linear Algebra II HW 10

Due Date: on or before 18:00, May/8/2025 (Thursday), in tutorial.

1. Show that if B is a real symmetric invertible matrix, then B^2 is positive definite.

2. Let $\begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.

(a) Show that A is positive definite and that $\boldsymbol{x}^T A\boldsymbol{x} = \boldsymbol{x}^T B\boldsymbol{x}$ for all $\boldsymbol{x} \in \boldsymbol{R}^2$.

(b) Show that B is positive definite, but B^2 is not positive definite.

3. For each of the following functions, find the stationary point and determine the stationary point corresponds to a local minimum, local maximum, or saddle point:

(a) $f(x,y) = \frac{y}{x^2} + \frac{x}{y^2} + xy$

(b) $f(x,y,z) = x^3 + xyz + y^2 - 3x$

4. For each of the following matrices, compute the determinants of all the leading principal submatrices and use them to determine whether the matrix is positive definite:

(a) $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

(b) $\begin{pmatrix} 3 & 4 \\ 4 & 2 \end{pmatrix}$

(c) $\begin{pmatrix} 6 & 4 & -2 \\ 4 & 5 & 3 \\ -2 & 3 & 6 \end{pmatrix}$

(d) $\begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & -2 \\ 1 & -2 & 5 \end{pmatrix}$

5. Let A be a symmetric $n \times n$ matrix and $\boldsymbol{x}$ be an $n \times 1$ column vector. Suppose $\boldsymbol{x}, A\boldsymbol{x}, \cdots, A^{n-1}\boldsymbol{x}$ are linearly independent and $A^n\boldsymbol{x} = -a_{n-1}A^{n-1}\boldsymbol{x} - \cdots - a_1 A\boldsymbol{x} - a_0\boldsymbol{x}$.

(a) Show that A is similar to $B = \begin{pmatrix} -a_{n-1} & \cdots & -a_1 & -a_0 \\ 1 & 0 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ \mathbf{0} & & 1 & 0 \end{pmatrix}_{n\times n}$.

(b) Suppose $1 + a_{n-1} + \cdots + a_1 + a_0 = 0$. Show that 1 is an eigenvalue. Find an eigenvector of A corresponding to eigenvalue 1.